

Hitachi Altaire PM Report



Document #: HMS102015

Revision: B

Affected Products

Altaire

Reference Material

N/A

History

Date	Name	Rev	Change Descriptions
10/14/15	Document Control	A	Initial release
2/22/16	Document Control	B	Reformatted sections, added coils data

Approval

Department	Name	Date
Product Specialist	Scott Mcpeek	10/14/15
QA & Safety	N/A	
Document Control	Jhovany Hernandez	2/22/16

Preface

This document is valid proof of a Preventative Maintenance performed on the system. The Preventative Maintenance must be performed in the specified intervals, and the results are to be documented herein.

This section addresses planned maintenance that should be performed by service personnel. Some of it is also listed in the Operator's Manual for the user to perform on daily and weekly basis.

However, service personnel should make sure that these procedures have been followed and if not, include them on the periodic visits.

This document must be completely filled out by the Field Service Engineer, i.e.:

- All boxes must be filled out. Use N/A where appropriate.
- Enter the Serial Number and the Date in the header of each page.
- After completing the check, fill out and add your name on the last page.

Equipment Needed

- Common tools

Serial #:

Date:

System Information

Customer:

Model:

System S/N:

System ID:

OEM System ID:

Measuring Equipment Used

Enter the measuring equipment used in the table below (meters, scopes, phantoms, etc.):

Equipment:

Serial #:

Serial #:

Date:

Checklist

Operational Checks

Pass	Fail	N/A	Action	Frequency*	Perform In
☐	☐	☐	Verify function of all switches on Gantry Control Panels	Monthly	Jan-Dec
☐	☐	☐	Verify Window Width and Level perform properly	Monthly	Jan-Dec
☐	☐	☐	Use FOV Phantom to Measure from Pins to Verify Measurement Function	Monthly	Jan-Dec

Magnet

Pass	Fail	N/A	Action	Frequency*	Perform In
☐	☐	☐	Visually check service port for ice blocks and contaminants	Annually	Jun
☐	☐	☐	Visually check Burst Discs for damage	Annually	Jun
☐	☐	☐	Check Joints for leaks	Annually	Jun
☐	☐	☐	Check relief valve for leaks	Annually	Jun
☐	☐	☐	Check all Seams of Cryostat for leaks	Annually	Jun
☐	☐	☐	Visually check Outdoor Exhaust Vent for blockage	Annually	Jun
☐	☐	☐	Check Compressor Gas Pressure is accurate	Annually	Jun
☐	☐	☐	Check Power Supply Conductor and Protective Earth Conductor for loose connections	Annually	Jun

*Unless otherwise specified by contract.

Serial #:

Date:

Gantry

Pass	Fail	N/A	Action	Frequency*	Perform In
☐	☐	☐	Check terminal connectors on Gantry for loose or damaged parts	Annually	Jan
☐	☐	☐	Visually check electrical components for damage on Gantry	Annually	Jan
☐	☐	☐	Check that Light Localizers function properly	Annually	Jul
☐	☐	☐	Clean RF covers	Annually	Jul
☐	☐	☐	Check Head Coil for damage to cable and case	Annually	Jul
☐	☐	☐	Check Flex QD Coils for damage to cable and case	Annually	Jul
☐	☐	☐	Check Joint Coils for damage to cable and case	Annually	Jul
☐	☐	☐	Check locks on each coil	Annually	Jul
☐	☐	☐	Check GC Cooling Hoses for leaks and damage	Annually	Jul

Table

Pass	Fail	N/A	Action	Frequency*	Perform In
☐	☐	☐	Check Indicator for Longitudinal movement	6-months	Feb/Aug
☐	☐	☐	Check distance in/out moved agrees with displayed value	6-months	Feb/Aug
☐	☐	☐	Verify Set moves table to center	6-months	Feb/Aug
☐	☐	☐	Verify that STOP button functions properly	6-months	Feb/Aug
☐	☐	☐	Check PCBs for damage	6-months	Feb/Aug
☐	☐	☐	Check Electrical components for damage	6-months	Feb/Aug
☐	☐	☐	Verify Power Supply Voltages	Annually	Aug
☐	☐	☐	Check Table Connector for damage	Annually	Aug
☐	☐	☐	Clean Tabletop	Annually	Aug
☐	☐	☐	Check for smooth Vertical travel of Table to its up/down limits	Annually	Aug
☐	☐	☐	Check for smooth Lateral travel of Table to its in/out and right/left limits	Annually	Aug

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Serial #:

Date:

IP/OP Unit

Pass	Fail	N/A	Action	Frequency*	Perform In
☐	☐	☐	Verify proper function of the Emergency button	6-months	May/Sep
☐	☐	☐	Clean Monitor	6-months	May/Sep
☐	☐	☐	Check and Clean Keyboard	6-months	May/Sep
☐	☐	☐	Check and Clean Mouse	6-months	May/Sep
☐	☐	☐	Check Intercom Function	6-months	May/Sep
☐	☐	☐	Check films print properly	6-months	May/Sep
☐	☐	☐	Verify buzzer operation at end of scan	6-months	May/Sep
☐	☐	☐	Clean and Check Fans	6-months	May/Sep
☐	☐	☐	Verify power supply voltages in Console(OP unit)	Annually	Sep
☐	☐	☐	Clean and Check Air Filters	6-months	May/Sep
☐	☐	☐	Check PCBs for damage	6-months	May/Sep
☐	☐	☐	Check for loose cables/connectors	6-months	May/Sep
☐	☐	☐	Clean and Check Fans	6-months	May/Sep
☐	☐	☐	Check terminal connectors in IP for loose or damaged parts	Annually	May
☐	☐	☐	Check Electrical components for damage	6-months	May/Sep
☐	☐	☐	Clean inside of IP cabinet	6-months	May/Sep
☐	☐	☐	Check operation of WS PCB using the "Check Program"	Annually	Sep
☐	☐	☐	Verify power supply voltages in IP unit	Annually	May
☐	☐	☐	Clean MOD Unit.	6-months	May/Sep

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Serial #:

Date:

RFDC

Pass	Fail	N/A	Action	Frequency*	Perform In
☐	☐	☐	Clean inside of RFDC cabinet	6-months	Apr/Oct
☐	☐	☐	Check Electrical components for damage	6-months	Apr/Oct
☐	☐	☐	Clean Air Filters	6-months	Apr/Oct
☐	☐	☐	Check for loose cables/connectors	6-months	Apr/Oct
☐	☐	☐	Check Intercom Function	6-months	Apr/Oct
☐	☐	☐	Check terminal connectors in RFDC for loose or damaged parts	Annually	Apr
☐	☐	☐	Verify power supply voltages in RF unit	Annually	Oct
☐	☐	☐	Verify power supply voltages in Shim/B0 unit	Annually	Oct
☐	☐	☐	Verify power supply voltages in DGBC/MPX unit	Annually	Oct

GCPA

Pass	Fail	N/A	Action	Frequency*	Perform In
☐	☐	☐	Clean inside of GCPA cabinet	6-months	Apr/Oct
☐	☐	☐	Clean Air Filters	6-months	Apr/Oct
☐	☐	☐	Check Electrical components for damage	6-months	Apr/Oct
☐	☐	☐	Clean and Check Fans	6-months	Apr/Oct
☐	☐	☐	Check for loose connectors	Annually	Apr
☐	☐	☐	Check terminal connectors, AC Input and DC Bus connections for loose/damaged parts	Annually	Apr

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Serial #:

Date:

EDRU

Pass	Fail	N/A	Action	Frequency*	Perform In
☐	☐	☐	Verify ERDU Function and Display	6-months	May/Nov
☐	☐	☐	Check Electrical components for damage	Annually	Nov
☐	☐	☐	Check for loose cables/connectors	Annually	Nov
☐	☐	☐	Replace Battery(Only every 2 years)	Annually	Nov
☐	☐	☐	Verify operation of the ERDU and ERDU circuit	Annually	Nov

SENS

Pass	Fail	N/A	Action	Frequency*	Perform In
☐	☐	☐	Check for leaks and damage	6-months	May/Nov
☐	☐	☐	Check for loose connections	Annually	Nov

Filter Box

Pass	Fail	N/A	Action	Frequency*	Perform In
☐	☐	☐	Clean inside of Filter Box	Annually	Nov
☐	☐	☐	Check Electrical components for damage	6-months	May/Nov
☐	☐	☐	Check terminal connectors in Filter Box for loose or damaged parts	Annually	Nov
☐	☐	☐	Tighten screws in Filter Box	Annually	Nov

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System Performance

Pass	Fail	N/A	Action	Frequency*	Perform In
☐	☐	☐	Verify and tune the transmitters(if needed)	6-months	May/Sep
☐	☐	☐	Verify FOV's and Check Polarity of each axis. Fill out FOV Accuracy Table	6-months	May/Sep
☐	☐	☐	HEAD COIL	Monthly	Jan-Dec
☐	☐	☐	KNEE COIL	6-months	Apr/Oct
☐	☐	☐	QUAD C-SPINE COIL	6-months	May/Nov
☐	☐	☐	WRIST COIL	6-months	Jan/Jul
☐	☐	☐	JOINT(S)	6-months	Apr/Oct
☐	☐	☐	JOINT COIL	6-months	Feb/Aug
☐	☐	☐	L. EXTREMITY COIL	6-months	Feb/Aug
☐	☐	☐	MED. FLEX COIL	6-months	Jan/Jul
☐	☐	☐	LRG. FLEX COIL	6-months	May/Sep
☐	☐	☐	XL. FLEX COIL	6-months	May/Sep
☐	☐	☐	MA PERIPHRAL VASCULAR	6-months	Jan/Jul
☐	☐	☐	MA BREAST	6-months	Feb/Aug
☐	☐	☐	MA NEUROVASCULAR	6-months	May/Sep
☐	☐	☐	MA SHOULDER	6-months	Apr/Oct
☐	☐	☐	MAC CTL	6-months	May/Nov
☐	☐	☐	MC CTL-SDQ	6-months	May/Nov
☐	☐	☐	Dual Quad	6-months	Jun/Dec
☐	☐	☐	TMJ	6-months	Jun/Dec
☐	☐	☐	RAPID HEAD	6-months	Apr/Oct
☐	☐	☐	RAPID BODY	6-months	May/Nov

Serial #:

Date:

System Performance (Cont'd.)

Pass	Fail	N/A	Action	Frequency*	Perform In
☐	☐	☐	RAPID EXTREMITY	6-months	May/Nov
☐	☐	☐	Verify Gating Functions Properly	6-months	Jun/Dec
☐	☐	☐	Verify O2 Monitor is functioning properly	6-months	Jun/Dec
☐	☐	☐	Verify Call Ball functions properly	6-months	Jun/Dec

Backup

Pass	Fail	N/A	Action	Frequency*	Perform In
☐	☐	☐	Perform Service Backup	Monthly	Jan-Dec
☐	☐	☐	Perform Protocol Backup	6-months	May/Sep

Call Bulb

Pass	Fail	N/A	Action	Frequency*	Perform In
☐	☐	☐	Check the hose kinks, knots, pinches, or tears	2-months	Feb/Apr/Jun/ Aug/Oct/Dec
☐	☐	☐	Check the connection of the hose line to the call bulb and the pressure switch	2-months	Feb/Apr/Jun/ Aug/Oct/Dec
☐	☐	☐	Verify the operation of the Call Bulb(replace batteries if needed)	2-months	Feb/Apr/Jun/ Aug/Oct/Dec

UPS

Pass	Fail	N/A	Action	Frequency*	Perform In
☐	☐	☐	Check indication light(s) for UPS functionality	6-months	Jun/Dec
☐	☐	☐	Clean and check Fans within UPS	6-months	Jun/Dec

Labels

Pass	Fail	N/A	Action	Frequency*	Perform In
☐	☐	☐	Verify that the Laser Warning Labels have been installed according to MRP 791	Annually	Jun

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Coil Data

<u>Head</u>	Gain	H1	SNR	
Actual				
Specification	N/A	<330	>=140	
<u>Knee</u>	Gain	H1	SNR	
Actual				
Specification	N/A	N/A	>=310	
<u>Quad C Spine</u>	Gain	H1	SNR	
Actual				
Specification	N/A	N/A	>=145	
<u>Wrist</u>	Gain	H1	SNR	
Actual				
Specification	N/A	N/A	>=290	
<u>Joint/Neck</u>	Gain	H1	SNR	
Actual				
Specification	N/A	N/A	>=120	
<u>L. Extremity</u>	Gain	H1	SNR	
Actual				
Specification	N/A	N/A	>=120	
<u>MA Shoulder</u>	Gain	H1	SNR (1)	SNR (2)
Actual				
Specification	N/A	N/A	>=75	>=160
<u>Medium Flex</u>	Gain	H1	SNR	
Actual				
Specification	N/A	N/A	>=85	
<u>Large Flex</u>	Gain	H1	SNR	
Actual				
Specification	N/A	N/A	>=100	
<u>XL Flex</u>	Gain	H1	SNR	
Actual				
Specification	N/A	N/A	>=85	

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System Condition

After performing scheduled PM, an evaluation of the system is performed with the following result (check one):

☐

No deficiencies. The image quality test resulted in no differences from required reference values.

☐

Slight deficiencies. There is no effect on operation; however they should be corrected preventively. The image quality test resulted in no differences from required reference values.

☐

Serious deficiencies. This must be corrected before operation of the system is permitted.

Field Service Engineer

Name:

Date:

Notes (Additional work to be performed, partial PM, customer requests):

Customer

Name:

Date: